SegMind: Semisupervised Remote Sensing Image Semantic Segmentation With Masked Image Modeling and Contrastive Learning Method

Zhenghong Li, Hao Chen, Jiangjiang Wu, Jun Li, and Ning Jing

***Abstract*— Remote sensing (RS) image semantic segmentation has attracted much attention due to its wide applications. However, deep learning-based RS image semantic segmentation methods usually require substantial manual pixelwise annotations, which are expensive and hard to obtain in practice. Although the existing semisupervised RS semantic segmentation methods effectively reduce dependence on labeled data, they generally focus on information consistency between labeled and unlabeled images, but ignore the potential context information between different areas of the RS image. In fact, the objects contained in an RS image usually have some long-range dependence between each other, since trees are usually on both sides of a road, and the middle of two rows of houses is commonly a road. Therefore, we believe that the potential dependencies between different areas of the RS image should be beneficial to reduce the label dependence of RS semantic segmentation. Based on this point, we propose a novel semisupervised RS image semantic segmentation network named SegMind, which is based on mean-teacher (MT) architecture and adopts masked image modeling (MIM) to enhance information interactions of different areas. Moreover, contrastive learning (CL) and entropy loss are introduced to SegMind framework to further improve the linear separability and prediction confidence of the proposed model. Experiments on three datasets have demonstrated the superiority of the proposed method over the state-of-the-art methods. The code is available at https://github.com/lzh-ggs-ddu/SegMind.**

***Index Terms*— Contrastive learning (CL), masked image modeling (MIM), remote sensing (RS) image semantic segmentation, semisupervised learning.**

I. Introduction

With the advent of remote sensing (RS) big data era, RS images develop explosively and have a wide range of applications [1]. RS image semantic segmentation, whose goal is to assign semantic labels to RS image pixels automatically, plays an important role in a broad range of applications, such as precision agriculture [2], urban planning [3], and land use analysis [4].

RS image semantic segmentation is challenging because of pixelwise prediction target and complex surface objects [5], [6]. Currently, deep learning methods have become the mainstream of RS image semantic segmentation. Compared with traditional methods, deep learning-based methods can automatically extract features and have demonstrated much superior performance on RS image semantic segmentation task [7]. Nevertheless, the superior performance of the deep learning-based methods relies heavily on a large amount of high-quality annotation data for training.

Although available RS images are numerous and image spatial resolution has increased a lot, pixelwise RS image labels are quite inadequate and hard to obtain in practice, because pixelwise annotations are labor-intensive, time-consuming, and expensive [8]. As [9] indicates, pixel-level annotations of one cityscapes image take almost 90 min on average. Another report [10] also notes that pixel-level annotations of DeepGlobe dataset need professional annotators but human errors are inevitable.

In order to mitigate the problem of lacking enough precise RS image labels, researchers mainly propose weakly supervised methods, unsupervised methods, and semisupervised methods. Weakly supervised methods [8], [11] only utilize lesser form of annotations (such as scribbles and points) instead of high-quality annotations. Although lesser form of annotations are easier to obtain than pixelwise annotations, weakly supervised methods still need many manual annotations [12]. Unsupervised methods can learn without annotations by clustering algorithm [13] or generative algorithm [14]. However, the accuracy of unsupervised methods is generally lower than that of supervised methods [15]. Semisupervised methods [16] employ a small labeled set and a large unlabeled set for training. The accuracy of semisupervised methods with much less labeled data can be very close to that of full-supervised methods with total labels [17], [18].

However, current semisupervised semantic segmentation methods generally focus on extracting the potential consistency information from both the labeled and unlabeled data effectively, ignoring the interactions between different areas of the RS image. Taking mean teacher (MT) as an example, the training process of the student module with unlabeled images is absolutely guided by the teacher module predictions. But,

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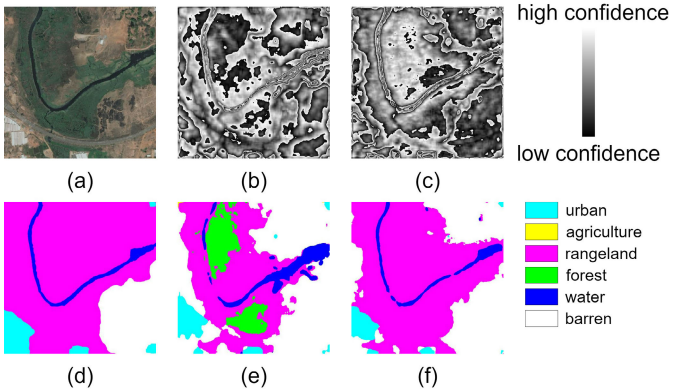


Fig. 1. Results on DGLCC dataset in 10% labeled rate with/without MIM. The network is MT with UNetFormer. (a) RS image. (b) Confidence map without MIM. (c) Confidence map with MIM. (d) Semantic segmentation label. (e) Semantic segmentation prediction without MIM. (f) Semantic segmentation prediction with MIM.

the predictions from the teacher module may be incorrect, thus misleading the training process of the student module. The combination of MT and pseudo-label approach [19] can overcome the problem to some extent, but pseudo-label approaches introduce a new problem that the information of the unreliable area of the RS image is not utilized. U2PL [20] combines MT, pseudo-label, and contrastive learning (CL) to use the latent information of the unreliable pseudo-label, but U2PL only adopts unreliable pixels as feature memory bank to lift pixel discriminative ability, and ignores the context information between different areas of the RS image. The objects contained in an RS image usually have some long-range dependencies between each other [21], [22]. For example, there are usually trees on both sides of a road, and the middle of the two rows of houses is a road. Therefore, we argue that the context information between different areas can also be utilized for semisupervised learning.

Recently, masked image modeling (MIM) has shown great success in the field of self-supervised learning [23]. MIM reconstructs masked areas from unmasked areas, which enhances image area information interactions and optimizes network weight without image labels [24]. As Fig. 1 shows, the confidence [the probability value of prediction class, mentioned in (2)] distribution with MIM is generally smoother, indicating better area information interactions with MIM. Considering river is slender and long range, the prediction results of Fig. 1(e) and (f) indicate that the MIM can utilize context information to improve semisupervised semantic segmentation performance.

Despite MIM has got great success in self-supervised learning, some researchers have pointed out that learned representations by MIM suffer from poor linear separability [23], [26]. CL methods focus on learning relations among features of different classes. CL can reduce the distance between similar samples and increases the distance between dissimilar samples in embedding space for both labeled and unlabeled data, which can enhance the discriminative capacity of MIM and lift the performance of a semisupervised learning model [27]. However, some researchers [28] point out that CL is sensitive to data augmentation and selection of negative pairs, which may hurt the prediction performance of semisupervised learning. We notice that MIM can play the role of data augmenters [29], and it does not require negative pairs, such as CL methods. Therefore, we believe that the combination of MIM and CL can make up each other's shortcomings in some extent.

As mentioned in aforementioned content, MIM and CL can utilize context information for semisupervised learning, CL can be beneficial to MIM by lifting the discrimination of representations, and MIM can be regarded as an additional data augmentation to enhance the performance of CL. In order to reduce the dependence of manual RS image annotations and to utilize the context information effectively, we propose a novel semisupervised remote sensing image semantic segmentation network based on masked image modeling and contrastive learning method, which is named SegMind. The contributions of this article are summarized as follows.

1) We combine MT framework with MIM and propose a novel semisupervised RS image semantic segmentation network named SegMind. To the best of our knowledge, it is the first work to introduce MIM into semisupervised RS image semantic segmentation task.
2) In order to further lift the performance of the proposed model, we make a new combination of MIM with CL. We only design two simple but effective projectors to generate compared features and reconstruct images, respectively, which can be effortless to apply in other networks.
3) We evaluate the proposed method on three RS datasets. The results show that the proposed SegMind outperforms several state-of-the-art semantic segmentation methods.

The remainder of this article is organized as follows. Section II introduces related work of semantic segmentation, semisupervised semantic segmentation, CL, and mask image modeling. The proposed method is illustrated in Section III. Experiments and analyses are presented in Section IV. Sections V and VI draw discussions and conclusions, respectively.

II. Related Work

In this section, we will first illustrate the concept and methods of semantic segmentation, thereafter semisupervised semantic segmentation, and, then, we will describe mask image modeling and CL methods.

A. Semantic Segmentation

Semantic segmentation refers to the process of assigning semantic labels to each pixels of an image, which is a basic task in computer vision. Before the popularity of deep learning, traditional semantic segmentation methods exploit contextual information based on manual designed features [30] [such as scale invariant feature transform (SIFT) and histograms of oriented gradients (HOG)] and traditional machine learning methods [31] (such as random forest and conditional random field). Because deep learning methods achieve remarkable performance, deep learning-based semantic segmentation methods have become the mainstream [7].

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Typical deep learning semantic segmentation models are mainly FCN [32], U-Net [33], SegNet [34], and DeepLab [35]. FCN is a revolutionary model, which replaces traditional fully connected layers with convolutional layers. U-Net is a U-shaped model with convolutional layers, deconvolutional layers, and a skip-connection structure. U-Net can get precise predictions with very few training images because of its elegant architecture. SegNet is based on an encoder–decoder structure, and its pooling operation in encoder retains max value places to upsample more precisely in decoder. DeepLab v3+ adopts atrous spatial pyramid pooling, depthwise separable convolution, and the encoder–decoder structure. It outperforms previous models in the PASCAL VOC 2012 dataset. Recently, transformer has widely applied in semantic segmentation. Segmenter [36] adopts pointwise linear mapping, which is based on a fully transformer-based encoder–decoder architecture. SegFormer [37] combines transformer and multilayer perceptron to avoid complex decoder. Recent RS image semantic segmentation methods are also influenced by transformer a lot. UNetFormer [38] contains convolutional neural network (CNN)-based encoder and transformer-based decoder, which captures both global and local contexts. ST-UNet [39] integrates transformer and CNN in parallel on encoder. MACU-Net [40] contains multiscale skip and asymmetric convolution based on U-Net.

B. Semisupervised Semantic Segmentation

Semisupervised learning was proposed to utilize a little labeled data and much unlabeled data. Semisupervised semantic segmentation methods can be divided into five classes [41].

1) *Adversarial Methods [42]:* Adversarial methods simulate real data distribution of both labeled and unlabeled data and, thereafter, generate new data or high-quality pseudo-labels by generative adversarial network (GAN) or variational auto-encoder (VAE).
2) *Consistency Regularization Methods [18], [43]:* Consistency regularization methods are based on manifold assumption or smoothness assumption, which tend to find a smooth manifold by unlabeled data. CCT [44] utilizes the consistency between main decoder prediction and auxiliary decoder prediction.
3) *Pseudo-Labeling Methods [45]:* Pseudo-labeling methods produce some high-quality pseudo-labels of unlabeled data to enhance the performance of the models iteratively. U2PL [20] develops pseudo-label methods by utilizing extra unreliable pseudo-label information.
4) *CL Methods [46]:* CL focuses on high-level features to differentiate between classes. CAC [47] compares features of the same area with different cropped operations based on CL.
5) *Hybrid Methods [48]:* Hybrid methods combine adversarial methods, pseudo-labeling, consistency regularization, entropy minimization, or other methods to enhance overall performance of model. For example, PSMT [49] combines generative methods and consistency regularization methods.

Semisupervised RS image semantic segmentation usually combines several semisupervised methods for better unlabeled image information mining ability. HTCR [50] mainly adopts consistency regularization to utilize unlabeled information. Yang et al. [51] proposed a novel semisupervised method combining pseudo-label and CL. Cui et al. [52] proposed a semisupervised method based on consistency and MT architecture.

C. Masked Image Modeling

MIM is a generative method that reconstructs original images from masked images [26]. Unlike masked language modeling widely adopted in natural language processing (such as GPT [53] and BERT [54]), MIM gets little attention in computer vision until recent years. The main reason of the difference is that earlier MIM methods [55], [56] perform worse compared with CL methods. BEiT [25] is the first MIM method that outperforms state-of-the-art contrastive method DINO [57].

MIM aims to acquire a general modeling representation ability by mask reconstruction. Masked autoencoder is a kind of MIM method that contains autoencoder [23], which is popular in recent years because of its outstanding performance. Self-attention layers of transformer provide the ability of global information interaction between patches [58]. MIM enforces the information interaction ability by masking and reconstructing process. Since masking can be seen as the process of adding noise, masked autoencoder can be seen as a form of denoising autoencoder [23]. MAE [59] only uses unmasked areas as encoder input and adopts mean-squared error loss. SimMIM [60] is a simple but effective end-to-end masked autoencoder, which adopts both masked and unmasked areas as encoder input. SemMAE [61] adopts semantic-guided mask strategy to integrate semantic information in the pretraining process. MIM is also applied in the field of RS. Gao et al. [62] verified the MIM masked model that focuses on key part of RS image. GSC-MIM [63] prevents the information loss of RS image local patches.

D. Contrastive Learning

Although CL in computer vision is popular in recent years because of its outstanding performance [64], the seminal idea traces back to the 1990s [65]. CL is a discriminative method that aims at learning to compare, i.e., to reduce the distance between similar samples and to increase the distance between dissimilar samples in embedding space [66]. The similarity between samples is usually evaluated by cosine similarity. The loss of CL is usually based on InfoNCE loss [67], which tends to maximize the lower bound of the mutual information between similar samples [68].

Recent years, CL in computer vision develops fast. MoCo [64] contains an encoder and a momentum encoder updated by exponential moving average (EMA). Previous batch features are maintained as a queue decoupled from mini-batch; therefore, the feature number of queue can be large. ReCo [46] adopts active hard sampling on queries and keys to reduce computing resources and memory resources. SimCLR [69] is a simple framework with a random augmentation module, an encoder, and a projection head. The method also points

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out that CL needs stronger data augmentation than supervised learning. BYOL [70] tends to get rid of negative samples by reducing the gap between different augmentations. CL is also adopted in RS semantic segmentation in recent years. GLCNet [71] applies contrastive loss in the end of encoder and decoder, thus achieving the global style CL and local feature CL. mCL-LC [72] introduces feature-level CL and semantic-level CL into RS image semantic segmentation.

III. Method

In this section, the architecture of SegMind will be illustrated first. Then, loss functions used in the proposed method will be formulated.

A. Architecture

The architecture of SegMind is illustrated in Fig. 2, which adopts MT as the semisupervised learning architecture. MT is based on consistency regularization, i.e., the prediction should not change after applying perturbations. The MIM can utilize the context information of RS imagery and play a role as a data augmenter to provide additional and helpful perturbations to MT, while the MT can effectively adjust the direction and magnitude of the gradients of MIM to enhance the prediction performance [73]. With the help of CL, the proposed model can do the consistency regularization not only in the prediction space but also in the feature space, leading to further increment of the discriminative ability of the proposed model. Therefore, SegMind is combined with MT, MIM, and CL methods.

As Fig. 2 shows, SegMind contains a teacher module and a student module with projector-r and projector-f. The teacher module and the student module adopt UNetFormer [38] as a backbone, which is shown in Fig. 3. Projector-f captures decoder features in order to compare feature difference. Projector-r is used to reconstruct images from encoder features. Projector-f and projector-r are convolution blocks, which are shown in Figs. 4 and 5, respectively.

Both of the teacher module and the student module contain ResBlock, global–local transformer block (GLTB), weight sum (WS) block, and feature refinement head (FRH) block. ResBlock is the block of ResNet. We replace ResNet18 in original UNetFormer encoder with ResNet50 for better feature extraction ability. GLTB captures global and local information. WS block fuses different features according to segmentation accuracy. FRH block shrinks the semantic gap between shallow features and high-level features. Projector-f is a shallow convolution block, which is adopted in previous contrastive methods [20], [46], [47]. Projector-r contains six convolution layers, which inputs encoder feature, masked image, and mask. Encoder feature and masked image in projector-r provide high-level and low-level context information for RS image reconstruction. Mask in projector-r provides position information for reconstruction.

The student module parameter θ_s is updated by label loss l_{lab}, entropy loss l_e, reconstruction loss l_r, reconstruction semantic consistency loss l_{rsc}, and contrastive loss l_c, which are formulated in Section III-C. The teacher module parameter is updated by EMA on student module parameter, i.e., the teacher module parameter θ_t is updated by

$$\theta_{t,j} = (1-\alpha_\theta)\theta_{t,j-1} + \alpha_\theta \theta_{s,j} \tag{1}$$

where $\theta_{t,j}$ is the j-iteration teacher module parameter, $\theta_{s,j}$ is j-iteration student module parameter, and α_θ is the hyperparameter of smoothing coefficient.

B. Training Process

The training set x is divided into labeled image x^l and unlabeled image x^u. Weakly augmented image x_{weak} and strongly augmented image x_{strong} are generated by weak augmentation and strong augmentation on x, respectively. Mask m is randomly generated according to mask rate. The size of m is the same as the size of x. The whole training process is shown in Algorithm 1.

Algorithm 1 Training Process of SegMind

Input: Labeled image x^l and with label y^l; Unlabeled image x^u; Iteraction number It; MIM iteraction number It_m; Loss weights α_e, α_r, α_{rsc} and α_c; Smoothing conefficient α_θ;
Output: Student model parameter θ_s and teacher model parameter θ_t;
Randomly initialize the model parameter θ_s and θ_t
$iternum$=0
for $iternum < It$ **do**
  Randomly generate mask m
  Input x^l and x^u into teacher model, then output prediction $\hat{y}_t$
  Get pseudo label y_u and teacher entropy map e_t according to $\hat{y}_t$
  Input x^l and x^u into student model, then output feature f, prediction $\hat{y}_s$ and entropy map e_s
  Calculate l_{lab}, l_e and l_c according to y_l, y_u, $\hat{y}_s$, e_s and f
  if $iternum < It_m$ **then**
    Input masked image into student model, then output reconstructed image $\hat{x}_{s,r}$ and prediction $\hat{y}_{s,m}$
    Calculate l_r and l_{rsc} according to $\hat{x}_{s,r}$, $\hat{y}_{s,m}$, x^l, x^u, y^l and y^U
    $l_{total} \leftarrow l_{lab} + \alpha_e l_e + \alpha_r l_r + \alpha_{rsc} l + rsc + \alpha_c l_c$
  else
    $l_{total} \leftarrow l_{lab} + \alpha_e l_e + \alpha_c l_c$
  end
  Update $\theta_{s,j}$ according to l_{total}
  $\theta_{t,j} \leftarrow (1-\alpha_\theta)\theta_{t,j-1} + \alpha_\theta \theta_{s,j}$
  $iternum$+=1
end
return θ_s, θ_t

First, x_{weak} and x_{strong} are sent into teacher module and student module, respectively. The unlabeled teacher module prediction $\hat{y}_t^u$ updates pseudo-label y^u by (2), i.e., the prediction class of high-confidence pixel is corresponding pixel pseudo-label. Label loss l_{lab} is based on the student module

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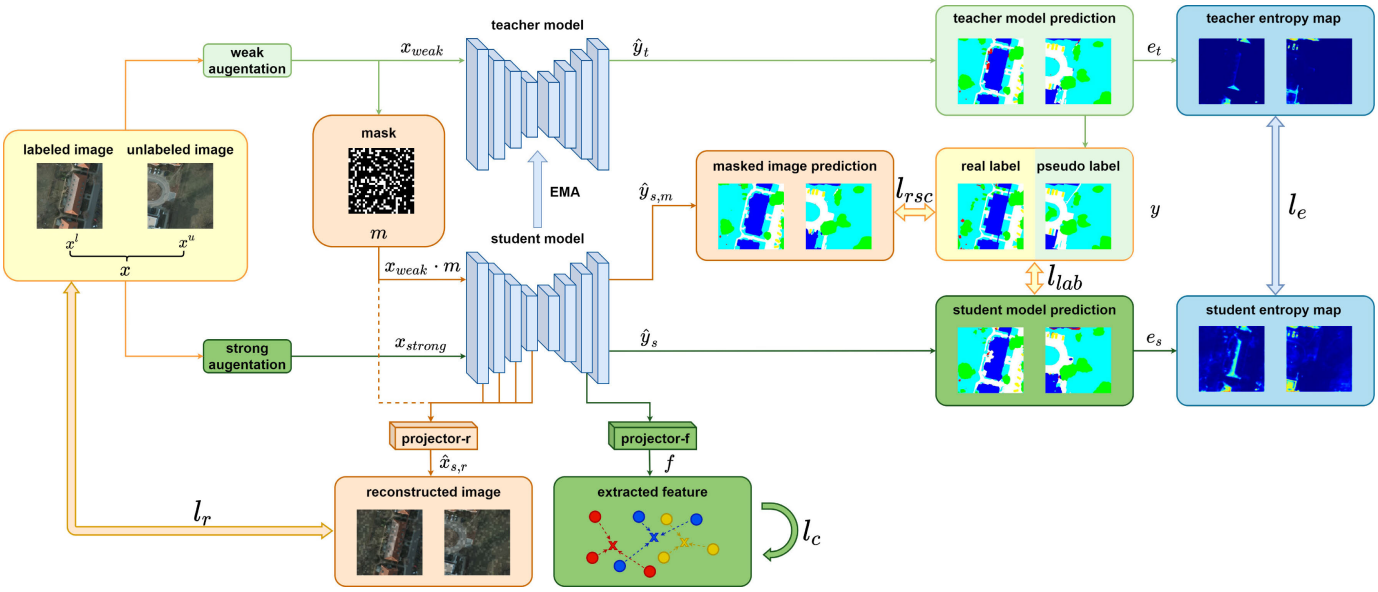


Fig. 2. Overview of the proposed SegMind network. SegMind contains a teacher module and a student module with projector-r and projector-f. Training image x generates weak augmentation image x_{weak} and strong augmentation image x_{strong} after weak augmentation and strong augmentation, respectively. x_{weak} and x_{strong} are sent to teacher module and student module, respectively; then, models generate predictions $\hat{y}_t$ and $\hat{y}_s$, respectively. Features in student module pass through projector-f and turn into extracted feature f. Pseudo-label is from unlabeled image prediction of teacher module. Teacher entropy map e_t and student entropy map e_s are calculated by teacher module prediction $\hat{y}_t$ and student module prediction $\hat{y}_s$, respectively. Masked image $x_{\text{weak}} \cdot m$ is generated by weakly augmented image x_{weak} and random mask m. Masked image is sent to student module, and projector-r then generates masked image prediction $\hat{y}_{s,m}$ and reconstructed image $\hat{x}_{s,r}$.

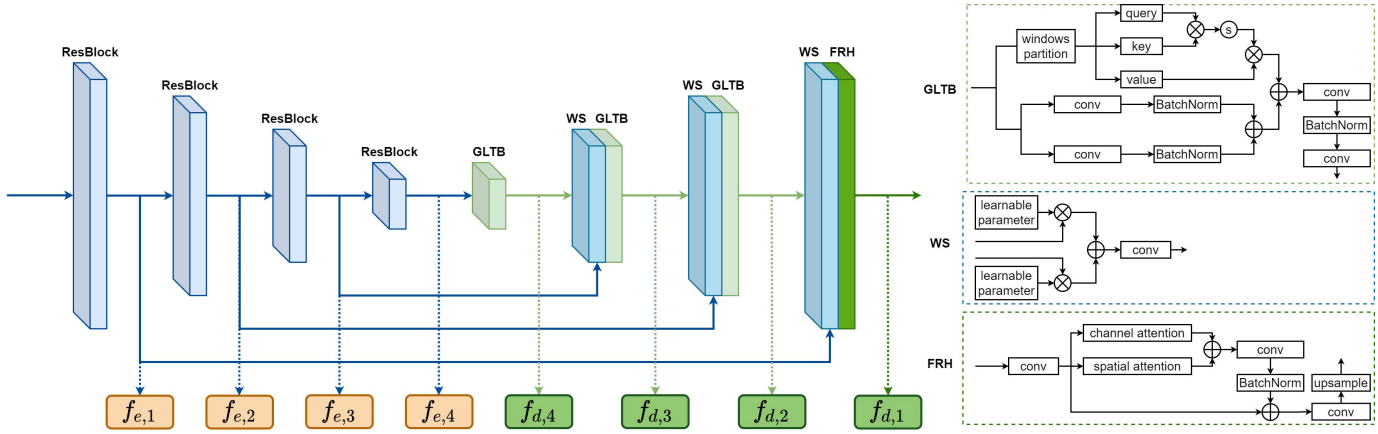


Fig. 3. Structure of UNetFormer. ResBlock is the block of ResNet50. GLTB represents global–local transformer block. WS represents weight sum block. FRH represents feature refinement block. ResBlock makes up UNetFormer encoder. GLTB, WS, and FRH make up UNetFormer decoder. $f_{e,1}$, $f_{e,2}$, $f_{e,3}$, and $f_{e,4}$ are features in UNetFormer encoder. $f_{d,4}$, $f_{d,3}$, $f_{d,2}$, and $f_{d,1}$ are features in UNetFormer decoder.

prediction $\hat{y}^u_{t,i}$ and label y, which utilizes labeled image information and unlabeled image information. The pseudo-label is calculated by

$$y^u_i = \begin{cases} \arg\max\left(\hat{y}^u_{t,i}\right), & \max\left(\hat{y}^u_{t,i}\right) > \delta_p \\ \text{none}, & \text{else} \end{cases} \tag{2}$$

where y^u_i is the ith pseudo-label pixel of unlabeled image x^u, $\hat{y}^u_{t,i}$ is teacher module prediction on the ith pixel of x^u, $\max(\hat{y}^u_t)$ is the confidence of the ith pixel, $\arg\max(\hat{y}^u_{t,i})$ is prediction class of $\hat{y}^u_{t,i}$, and δ_p is pseudo-label threshold.

Second, the entropy map e is generated from teacher module prediction $\hat{y}_t$ and student module prediction $\hat{y}_s$. Since lower uncertainty means that unreliable pseudo-labels are less and available pseudo-label information is more, decreasing prediction uncertainty is beneficial to improve performance [75]. The entropy map e reflects the uncertainty extend of prediction. Unlike [75] pushing all pixel entropy values close to 0, we notice area discrimination difference. Thus, designed entropy loss l_e decreases student module prediction uncertainty based on teacher entropy map e_t and student entropy map e_s. The entropy map e is calculated by

$$e = -\sum_{j=1}^{c} \hat{y}_j \log(\hat{y}_j) \tag{3}$$

where $\hat{y}_j$ denotes the probability of $\hat{y}$ to be class j, $\hat{y}$ is student module prediction $\hat{y}_s$ or teacher module prediction $\hat{y}_t$, e is student module entropy map e_s or teacher module entropy map e_t, and c is class number.

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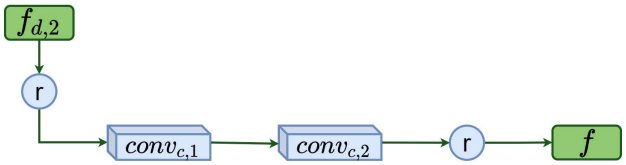


Fig. 4. Structure of projector-f. Resize operation is denoted as "r."

Then, projector-f extracts features f from student module decoder features $f_{d,2}$. Contrastive loss l_c reduces distance between the same class and increases distance between different classes, thus improving network linear separability. The structure of projector-f is shown in Fig. 4. Contrastive loss l_c is based on extracted features f and corresponding labels y.

After that, projector-r reconstructs image from mask m, masked image $x_{\text{weak}} \cdot m$, and student module encoder features f_e. Since reconstructed image masked area is generated by original image unmasked area information, reconstructing masked areas encourages information interactions between different areas. Mask m is randomly generated with the same size of x_{weak}, and its value is 0 (masked) or 1 (unmasked). Mask m provides position information for projector-r. The structure of projector-r is shown in Fig. 5. Masked area of reconstructed image $\hat{x}_{s,r}$ is compared with original image x by reconstructed loss l_r.

Finally, masked image $x_{\text{weak}} \cdot m$ is sent into student module. Similar to previous self-supervised MIM methods, l_r only encourages reconstructing original image. Note that label and pseudo-label in semisupervised learning can provide extra information, we further encourage area interactions in a semantic level. Since information of masked area is missing, l_{rsc} enhances connection of different areas in semantic level and lifts network feature extraction ability. Reconstruction semantic consistency loss l_{rsc} is based on masked image prediction $\hat{y}_{s,m}$ and corresponding label y.

C. Loss Function

Assume the height and width of images are H and W, respectively. Our method contains five losses: label loss l_{lab}, entropy loss l_e, reconstruction loss l_r, reconstruction semantic consistency loss l_{rsc}, and contrastive loss l_c. Total loss l_{total} is defined as follows:

$$l_{\text{total}} = l_{\text{lab}} + \alpha_e l_e + \alpha_r l_r + \alpha_{\text{rsc}} l_{\text{rsc}} + \alpha_c l_c \tag{4}$$

where α_e, α_r, α_{rsc}, and α_c are the weights of corresponding losses.

1) Label Loss: Following precious semantic segmentation works [76], we adopt cross-entropy loss on predictions, labels, and pseudo-labels. Label loss l_{lab} introduces high-confidence pseudo-label pixels into training, hence utilizing unlabeled information. Label loss l_{lab} is defined as follows:

$$l_{\text{lab}} = \frac{1}{n} \sum_{i=1}^{n} y_i \log(\hat{y}_i) \tag{5}$$

where y_i is the label or pseudo-label of the ith pixel. $\hat{y}_i$ is the prediction of the ith pixel, and n is the number of pixels participating in calculation.

For labeled images, all image pixels participate in calculation; therefore, n is equal to $H \times W$. For unlabeled images, only pixels whose confidence values higher than the threshold θ_p participate in calculation; therefore, n is equal to pseudo-label pixel number.

2) Entropy Loss: RS image usually covers large area and contains many objects, leading to complex boundaries. Previous studies [11], [77] have found that predictions on boundary areas tend to be low confidence (as shown in Fig. 1), which means tending to be with high-entropy values. Some semisupervised semantic segmentation methods [20], [78] have introduced entropy in training. Among them, [78] notes the entropy map difference between labeled image prediction and unlabeled image prediction, thus using an extra discriminator to prompt unlabeled prediction entropy maps similar to labeled prediction entropy maps. Inspired by above works and notice that teacher module predictions tend to be more accurate [79], we design entropy loss l_e based on the prediction entropy maps difference between teacher module and student module. Designed entropy loss l_e is defined as follows:

$$l_e = \frac{1}{H \times W} (e_t - e_s)^2 \tag{6}$$

where e_t and e_s are entropy maps of teacher module and student module, respectively.

3) Reconstruction Loss: Since RS image usually covers large area, utilizing context information can improve network discriminative ability in some extent [21]. As [24] points out, MIM uses unmasked areas to reconstruct masked areas, therefore forcing model to learn the interactions between different areas. In order to better extract context information from both labeled images and unlabeled images, we design reconstruction loss l_r based on MIM.

Combining [59] and [60], we put masked image $x_{\text{weak}} \cdot m$ into student module and get reconstructed image $\hat{x}_{s,r}$, and then calculate the difference between reconstructed image $\hat{x}_{s,r}$ and original image x_{weak} by mean-squared error loss. Mask m only contains two different values, where 1 represents unmasked area and 0 represents masked area. Reconstruction loss is defined as follows:

$$l_r = \frac{1}{\mathbb{1}(m=0)} (\hat{x}_{s,r} \cdot \mathbb{1}(m=0) - x_{\text{weak}} \cdot \mathbb{1}(m=0))^2 \tag{7}$$

where $\hat{x}_{s,r}$ is reconstructed image of student module, x_{weak} is original images with weak augmentation, $\mathbb{1}(\cdot)$ is the indicator function, and $\mathbb{1}(m=0)$ represents masked area, i.e., the pixels that correspond mask values equal to 0.

4) Reconstruction Semantic Consistency Loss: Since MIM is mainly applied in self-supervised learning to pretrain without labels, earlier MIM methods [25], [59], [60] tend to reconstruct original images but ignore reconstructed image semantic information. Later MIM methods [29] use semantic information to adjust mask strategy but ignore segmentation information that can guide network to update parameter. Since semisupervised semantic segmentation is a dense prediction task, the high-level features are quite important [80]. In order to enhance semanticwise interaction of different areas, we utilize label information and propose reconstruction semantic

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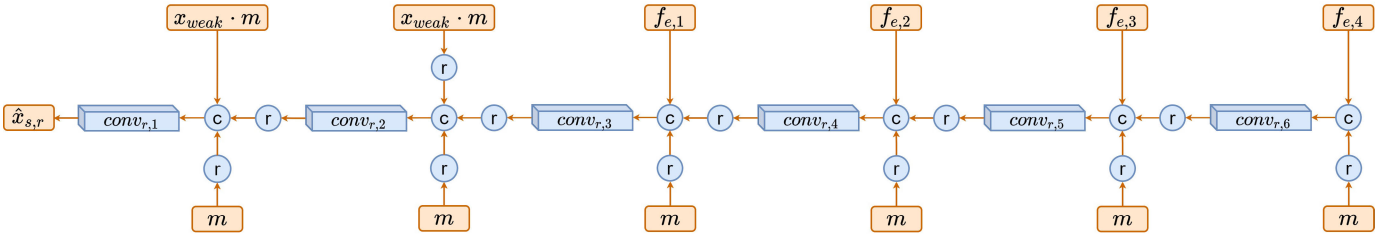


Fig. 5. Structure of projector-r. Resize and concatenate operations are denoted as "r" and "c," respectively.

consistency loss l_{rsc}. Unlike l_r pushes reconstructed image $x_{s,r}$ similar to original image x_{weak} in pixel value level, l_{rsc} pushes masked image predictions in masked area similar to labels in masked area. Reconstruction semantic consistency loss l_{rsc} is defined as follows:

$$l_{\text{rsc}} = \frac{1}{\mathbb{1}(m=0)}(y \cdot \mathbb{1}(m=0)) \log(\hat{y}_{s,m} \cdot \mathbb{1}(m=0)) \quad (8)$$

where y is the label and $\hat{y}_{s,m}$ is the semantic segmentation prediction on $x_{\text{weak}} \cdot m$.

5) Contrastive Loss: Despite MIM enhances information iterations between different areas, it suffers from poor linear separability of learned representations [26]. In order to improve learned representation linear separability, we introduce contrastive loss l_c. It is defined as follows:

$$l_c = \sum_{f_q \in F_q} -\log \frac{e^{f_q \cdot f_k^{j,+}/\tau}}{e^{f_q \cdot f_k^{j,+}/\tau} + \sum_{f_k^{j,-} \in F_k^{j,-}} e^{f_q \cdot f_k^{j,-}/\tau}} \quad (9)$$

where F_q is query feature set, f_q is query feature, j is the class of query feature, $F_k^{j,-}$ is negative key feature memory bank of class j, $f_k^{j,-}$ is negative key feature of class j, $f_k^{j,+}$ is positive key feature of class j, and τ is temperature parameter that controls the softness of feature distribution.

We adopt mean feature of class j in the batch as positive key feature $f_k^{j,+}$. In order to maintain the balance between negative key feature number and memory resource, we build a dynamic queue followed [64]. The negative key feature queue $F_k^{j,-}$ represents all data in the queue except class j. Since sample strategy is important to CL [28], we sample query feature f_q that corresponding pixel confidence is lower than query threshold δ_q. Negative key number of current batch is according to the mean feature similarity between batch and queue.

IV. Experiment

In this section, we will introduce the datasets used in the experiment first. Then, we will introduce comparative methods and evaluation metrics used in the experiment. In Section IV-C, we will give a detailed description of training. In Section IV-D, we will display and analyze experiment results.

A. Dataset

To verify the effectiveness of the proposed method, we select three RS semantic segmentation datasets for experiment: LoveDA dataset [81], DeepGlobe Land Cover Classification (DGLCC) dataset [10], and Potsdam dataset [82].

TABLE I
Number of Dataset Images Used in the Experiment

Dataset	Training	Validation	Test	Total
LoveDA	2522	834	835	4191
DGLCC	2252	480	480	3212
Potsdam	1008	180	180	1368

1) *LoveDA Dataset [81]:* The dataset contains 5987 satellite images with the size of 1024 × 1024 pixels. The dataset covers 536.15-km^2 area with a resolution of 0.3 m/pixel. The pixel labels are divided into eight classes: background, building, road, water, barren, forest, agriculture, and no data. The no-data class is set as ignored class. Because original test set does not contain corresponding labels, we split original validation set into validation set and test set.
2) *DGLCC Dataset [10]:* The dataset contains 1146 satellite images with the size of 2448 × 2448 pixels. The dataset covers 1716.9-km^2 area with a resolution of 0.5 m/pixel. The pixel labels are divided into seven classes: urban, agriculture, rangeland, forest, water, barren, and unknown. In the experiment, we set unknown class as ignored class. Considering original validation and test sets are without labels, we split original training set into training, validation, and test sets. Besides, images are cropped with a size of 1024 × 1024.
3) *Potsdam Dataset [82]:* The dataset contains 38 satellite images with the size of 6000 × 6000. The labels are divided into six classes: impervious surface, building, low vegetation, tree, car, and clutter. In the experiment, we randomly split 38 images into training set (28 images), validation set (five images), and test set (five images). Images are then cropped with the size of 1024 × 1024.

The training set image number and labeled rate are denoted as n and λ, respectively. Training set is randomly split into labeled images and unlabeled images. The numbers of labeled images and unlabeled images are $\lambda \times n$ and $(1-\lambda) \times n$, respectively. In the experiment, labeled rates are set as 5%, 10%, and 20%. The number of dataset images is shown in Table I.

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B. Baselines and Metrics

To verify the effectiveness of the proposed method, some state-of-the-art RS image semantic segmentation methods and semisupervised semantic segmentation methods are selected as baselines, which are described as follows.

1) *UNetFormer [38]:* UNetFormer is an RS image semantic segmentation method with CNN encoder and transformer decoder. The backbone of the proposed model is based on UNetFormer.
2) *ST-UNet [39]:* ST-UNet is an RS image semantic segmentation method with transformer branch. The spatial interaction module of ST-UNet focuses pixel-level feature correlation.
3) *MACU-Net [40]:* MACU-Net is a CNN-based RS semantic segmentation method. The multiscale skip connected and asymmetric convolution of MACU-Net designed for fine-resolution RS images.
4) *ReCo [46]:* ReCo is a regional-level CL framework with active sampling. The method samples hard negative points with little additional memory footprint. It achieves smooth segmentation boundaries and fast convergence.
5) *U2PL [20]:* U2PL is a semisupervised semantic segmentation framework putting unreliable pseudo-labels into training. The method is based on that unreliable pseudo-labels can provide information about dissimilar classes. It makes full use of unreliable pseudo-label information by an additional classwise negative memory bank.
6) *CAC [47]:* CAC is a pixel-level CL method, which aims to maintain context-aware consistency. It adopts directional contrastive loss using overlap of different crops in the same image, which makes models robust to the contextual variance.
7) *CCT [44]:* CCT is a cross-consistency semisupervised semantic segmentation framework based on cluster assumption. Because low-density regions are more apparent within the hidden representations than within the inputs, the network adds additional auxiliary decoders to get different perturbations.
8) *CPS [83]:* CPS is a simple but effective consistency regularization approach. It imposes the consistency of two models with different initialization. Model parameters are updated alternately. The pseudo-label that updates one model is generated from the other model prediction.
9) *PSMT [49]:* PSMT is a semisupervised semantic segmentation method based on virtual adversarial training and consistency learning. The method uses confidence weighted cross entropy and a new auxiliary teacher to reduce inaccurate predictions.

In the experiment, we adopt three evaluation metrics to compare the performance of different methods: mean intersection over union (mIoU), $F1$ score ($F1$), and Kappa coefficient (Kappa). The evaluation metrics are formulated as follows:

$$\text{mIoU} = \frac{1}{c}\sum_{j=1}^{c}\frac{\text{TP}_j}{\text{TP}_j + \text{FP}_j + \text{FN}_j} \tag{10}$$

$$F1 = 2 \times \frac{\text{Pr} \times \text{Rec}}{\text{Pr} + \text{Rec}} \tag{11}$$

$$\text{Kappa} = \frac{\text{Po} - \text{Pe}}{1 - \text{Pe}} \tag{12}$$

where c, TP, FP, TN, and FN are class number, true positive, false positive, true negative, and false negative, respectively. Pr, Rec, Po, and Pe are defined as follows:

$$\text{Pr} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

$$\text{Po} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{15}$$

$$\text{Pe} = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{TN} + \text{FN})(\text{TN} + \text{FP})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{16}$$

C. Implementation Details

The experiments are implemented with PyTorch framework on a single NVIDIA RTX 3080 GPU. We adopt AdamW as an optimizer. Learning rate and weight decay of optimizer are set as 2×10^{-4} and 1×10^{-4}, respectively. We adopt polynomial learning rate decay: $(1 - (\text{iter}/\max_\text{iter}))^{0.9}$. Training epoch number and batch size are 100 and 2, respectively. l_r and l_{rsc} are applied in the first 50 epochs. The temperature in l_c is set as 0.5. The labeled rates of dataset are set as 0.05, 0.1, and 0.2. During training, we adopt classmix augmentation. Query threshold δ_q and pseudo-label threshold δ_p are 0.97 and 0.7, respectively. The mask rate and mask size are set as 0.25 and 4×4, respectively. Loss weights α_e, α_r, α_{rsc}, and α_c are set as 1, 1, 1, and 1, respectively.

The weak augmentation contains random crop, random flip, and random rotation. The strong augmentation includes color jitter, random gray scale, and random Gaussian blur in addition to weak augmentation. In color jitter process, variation range for brightness, contrast, saturation, and hue are 0.05, 0.05, 0.05, and 0.1, respectively. Probability for random gray scale and random Gaussian blur are 0.2 and 0.5, respectively. Images are resize to 512×512 before inputting into network.

D. Experimental Result

To verify the effectiveness of the proposed method, extensive experiment results are summarized and analyzed.

1) LoveDA: From Table II, it can be seen that SegMind outperforms other methods with different labeled rates. We can conclude that the proposed SegMind achieves the best performance against other methods. SegMind gets the highest mIoU, $F1$, and Kappa values in the dataset. Compared with ReCo, SegMind gets 2.55%, 2.33%, and 2.12% improvements in mIoU when labeled rates are 5%, 10%, and 20%, respectively. ReCo adopts batchwise contrastive loss rather than dataset-wise contrastive loss in SegMind. Compared with the U2PL, SegMind gets 6.67%, 6.66%, and 7.84% improvements of mIoU, $F1$, and Kappa, respectively, when the labeled rate is 5%. The improvements are 6.96%, 6.70%, and 6.93%, respectively, when the labeled rate is 10%. When the labeled rate increases to 20%, the improvements are 4.14%, 3.51%, and 4.12%, respectively. The improvements increase with the labeled rate decreasing, because MIM in SegMind can mine context information effectively without label. Besides, CAC,

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TABLE II
SUMMARY OF THE QUANTITATIVE RESULTS ON THE LOVEDA DATASET

Method	5%			10%			20%			100%		
	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa
UNetFormer [38]	0.4608	0.6248	0.5393	0.4647	0.6199	0.5630	0.4832	0.6421	0.5734	0.5230	0.6775	0.6112
ST-UNet [39]	0.2853	0.4369	0.3232	0.3203	0.4801	0.3543	0.3565	0.5215	0.4004	0.4369	0.6012	0.4953
MACU-Net [40]	0.2252	0.4060	0.3405	0.2657	0.4535	0.3768	0.3099	0.4569	0.4027	0.4416	0.6010	0.5270
ReCo [46]	0.4544	0.6124	0.5485	0.4824	0.6391	0.5772	0.4933	0.6524	0.5896	-	-	-
U2PL [20]	0.4132	0.5800	0.4795	0.4361	0.5977	0.5200	0.4731	0.6359	0.5533	-	-	-
CAC [47]	0.4052	0.5684	0.4932	0.4482	0.6106	0.5482	0.4470	0.6073	0.5373	-	-	-
CCT [44]	0.2926	0.4954	0.4227	0.3262	0.5443	0.4400	0.3931	0.5466	0.4454	-	-	-
CPS [83]	0.4365	0.5990	0.5153	0.4464	0.6072	0.5353	0.4487	0.6080	0.5268	-	-	-
PSMT [49]	0.2836	0.4047	0.3857	0.4475	0.6072	0.5463	0.4550	0.6155	0.5497	-	-	-
SegMind	**0.4799**	**0.6466**	**0.5579**	**0.5057**	**0.6647**	**0.5893**	**0.5145**	**0.6710**	**0.5945**	-	-	-

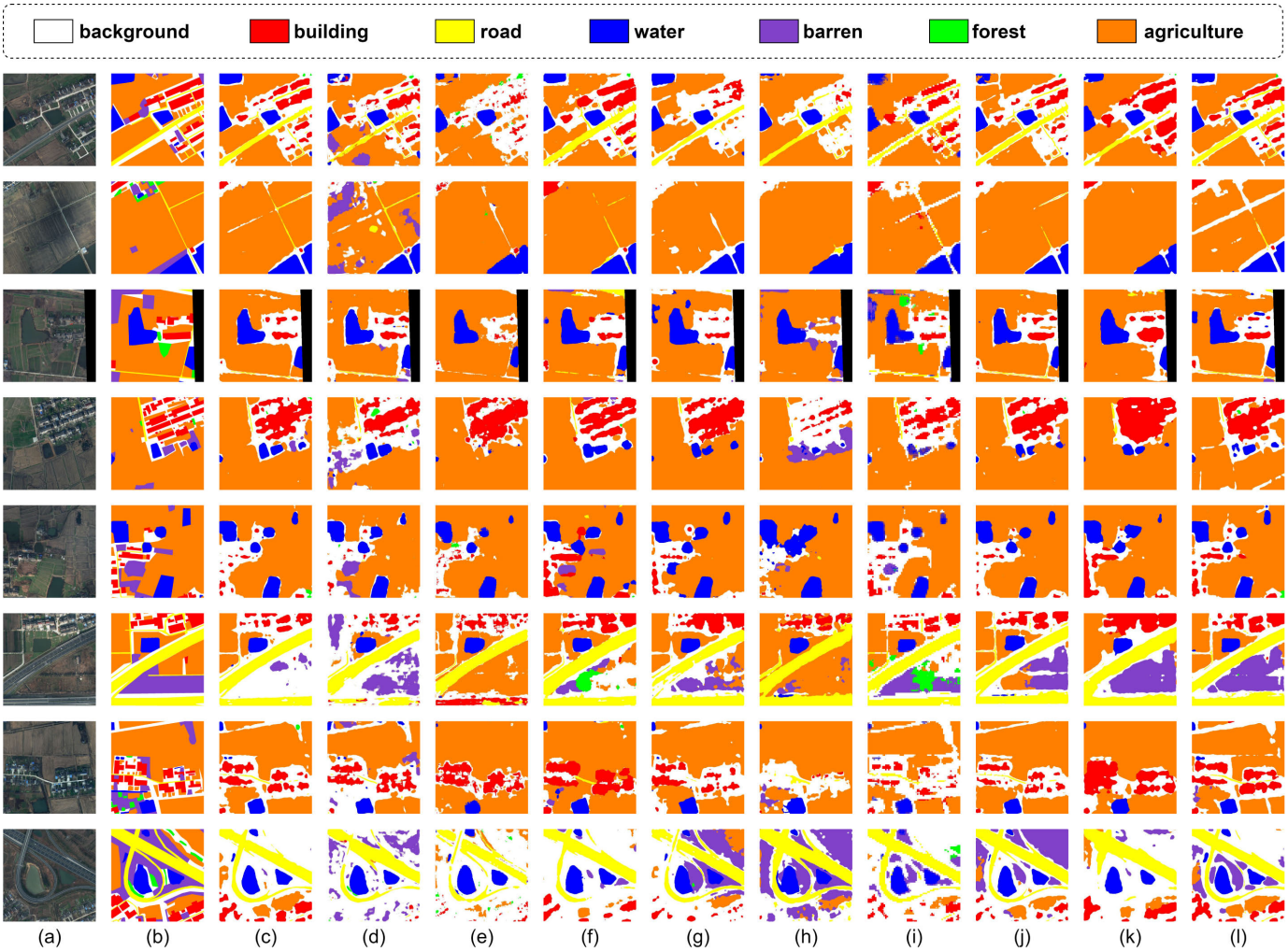


Fig. 6. Visual comparisons of different methods on the LoveDA dataset with 10% labeled rate. (a) RS image. (b) Ground truth. (c) UNetFormer. (d) ST-UNet. (e) MACU-Net. (f) ReCo. (g) U2PL. (h) CAC. (i) CCT. (j) CPS. (k) PSMT. (l) SegMind.

CCT, CPS, and PSMT get lower performance in mIoU, $F1$, and Kappa with 5%, 10%, and 20% labeled rates. Notably, CCT gets worst performance with 10% and 20% labeled rates, and PSMT gets worst performance with 5% labeled rate. Since RS contains many complex objects, perturbations in CCT aux decoder and high-level perturbations in PSMT worsen the recognition of objects.

In order to see the improvements more visually, we show some prediction results of different methods with 10% labeled rate in Fig. 6. From Fig. 6, we can observe that there are more

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missed false areas in the comparative methods. As the first row shows, ReCo produces rough outline but losses details (road in prediction is disconnected), and building in U2PL prediction is inaccurate. In the second row, U2PL and CAC loss red area compared with the ground truth, whereas the proposed SegMind generally achieves the best visual performance, and its results are much closer to the ground truth.

2) DGLCC: The quantitative evaluation results of different methods are shown in Table III. As the table shows, the proposed SegMind gets the highest $F1$, mIoU, and Kappa values against other methods in different labeled rates. ReCo gets 4.59% lower mIoU, 4.85% lower $F1$, and 1.51% lower Kappa than SegMind with 5% labeled rate. Compared with other baselines, ReCo gets high performance with low labeled rate. ReCo adopts contrastive loss and active hard sampling strategy, so that its linear separability is high. Besides, CCT and PSMT get low mIoU values with low labeled rate. CPS also achieves quite good results, and it achieves the second highest mIoU and $F1$ with 5%, 10%, and 20% labeled rate. When labeled rate reduces from 20% to 5%, the mIoU values of ReCo, U2PL, CAC, CCT, CPS, and PSMT reduce 10.45%, 10.65%, 9.39%, 18.39%, 10.62%, and 38.20%, respectively, SegMind only reduce 7.73%. Besides, SegMind reduces least, and U2PL reduces least in $F1$ and Kappa when labeled rate reduces from 20% to 5% in the DGLCC dataset. Since SegMind combines CL and MIM, it can effectively use the information of unlabeled images, and SegMind performance generally reduces less with labeled rate reducing.

In order to show performances visually, different methods results with 10% labeled rate are shown in Fig. 7. We can observe that large amount of false areas are in the comparative methods. The proposed SegMind achieves the best visual performance, and its results are closer to the ground-truth images. As the first row of the figure shows, U2PL, CAC, CPS, and PSMT results do not contain river. ReCo and CCT results in the first row contain river, but the outlines are less similar to ground truth than SegMind. In the second row, pink area in SegMind is the most similar to the ground truth. In the forth row, only SegMind recognizes the lake. In the fifth row, only CCT and SegMind recognize the right river. From the above, SegMind has better contextual information extraction capability. Overall, it can be seen from Fig. 7 that our semisupervised model outperforms the other baselines on the DGLCC dataset.

3) Potsdam: Table IV reports the quantitative evaluation results of different methods in different labeled rates. SegMind gets the highest $F1$, mIoU, and Kappa values. Especially, in the case of 5% labeled rate, SegMind achieves 67.42%, 78.13%, and 81.62% in mIoU, $F1$, and Kappa, respectively. The quite high performance in low labeled rate indicates that SegMind can make full use of unlabeled image information. SegMind is superior in accuracy, with 2.63%–36.22% higher mIoU, 3.01%–26.07% higher $F1$, and 1.15%–29.40% higher Kappa than the other methods. When labeled rate reduces from 20% to 5%, mIoU, $F1$, and Kappa of SegMind reduce the least among the seven methods. In the case of 5% labeled rate, the mIoU values of ReCo, U2PL, CAC, CCT, CPS, and PSMT are 9.87%, 9.32%, 14.63%, 25.26%, 7.71%, and 36.22% lower than SegMind, respectively. The higher performance with low labeled rate indicates that SegMind can utilize unlabeled information to optimize network. Unlike LoveDA dataset and DGLCC dataset, CPS results (mIoU, $F1$, and Kappa) are the second highest among the seven methods. Potsdam dataset images are less, but object details are more complex than LoveDA and DGLCC. The result indicates that cross pseudo-supervision in CPS is also good at distinguishing little and complex objects.

The results of different methods with 10% labeled rate are shown in Fig. 8. Many missed areas and false areas are in the comparative methods, and the proposed SegMind achieves the best visual performance. Taking the first row as an example, red area is missed in U2PL, CCT, and PSMT. Blue area is partly missed in ReCo, U2PL, CAC, CCT, CPS, and PSMT. SegMind result keeps red area and blue area; hence, the SegMind prediction is similar to the ground truth. In the second row, CPS and SegMind predictions keep more accurate small objects than other methods. In the sixth row, the parterre shape keeps well in CPS and SegMind. This can indicate that SegMind can keep object outlines better in some cases. In the seventh row, SegMind keeps some red areas as the ground truth, but predictions in other methods lack red area. Above observations indicate that SegMind generally achieves better visual performance compared with other methods.

The proposed method mainly contains three parts: MT, MIM, and CL. In order to verify the effectiveness of each component of the proposed SegMind, we conducted ablation experiments, and the results is shown in Table V. The experiment is conducted on tree datasets with 10% labeled rate. The first line in the table represents training without MT, MIM, or CL, i.e., training without unlabeled data. As the table shows, the introduction of MT, MIM, and CL improves performance. Besides, the last line in the table gains the highest performance. The result verifies the effectiveness of combing MT, MIM, and CL.

V. Discussion

In order to validate the effects of the proposed method and explore the influence of several parts in SegMind, we make some experiments and discuss the influence of these parts. We adopt the above datasets, and the labeled rate is set as 10%.

The experiment result about the loss function influence is shown in Table VI. We can conclude that l_e, l_r, l_{rsc}, and l_c are conducive to improve network performance. Result in Table VI indicates that l_e, l_r, l_{rsc}, and l_c improve 1.00%–1.70%, 0.81%–1.25%, 0.90%–2.10%, and 0.80%–1.77% mIoU values in the datasets, respectively. Although the mechanism of these loss functions is different, mIoU improvements of using l_e, l_r, l_{rsc}, and l_c are generally close. In order to explore how they improve network performance, we conduct some experiments to show the impact of loss functions.

We conduct experiments on the LoveDA dataset with 10% labeled rate to explore the influence of l_e. The result is shown in Fig. 9. Fig. 9(c) and (e) shows the heat maps of prediction entropy maps, redder color represents higher entropy value, and bluer color represents lower entropy value. Comparing

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TABLE III
SUMMARY OF THE QUANTITATIVE RESULTS ON THE DGLCC DATASET

Method	5%			10%			20%			100%		
	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa
UNetFormer [38]	0.5497	0.6827	0.6518	0.6174	0.7445	0.7172	0.6567	0.7801	0.7489	0.6943	0.8078	0.7864
ST-UNet [39]	0.3791	0.5119	0.5028	0.4547	0.5967	0.5888	0.5057	0.6498	0.6344	0.6260	0.7572	0.7360
MACU-Net [40]	0.3658	0.4717	0.5119	0.4116	0.6119	0.5829	0.5322	0.6483	0.6504	0.6205	0.7386	0.7281
ReCo [46]	0.5566	0.6865	0.6619	0.5842	0.7125	0.6985	0.6611	0.7760	0.7696	-	-	-
U2PL [20]	0.5100	0.6388	0.6536	0.6092	0.7444	0.7146	0.6165	0.7492	0.7066	-	-	-
CAC [47]	0.5473	0.6901	0.6219	0.6100	0.7445	0.7031	0.6412	0.7696	0.7371	-	-	-
CCT [44]	0.4685	0.7036	0.6345	0.6349	0.7638	0.7368	0.6524	0.7787	0.7540	-	-	-
CPS [83]	0.5523	0.6943	0.6622	0.5997	0.7340	0.7064	0.6595	0.7820	0.7577	-	-	-
PSMT [49]	0.2322	0.6273	0.3631	0.5322	0.6685	0.6428	0.6142	0.7435	0.7074	-	-	-
SegMind	**0.6025**	**0.7350**	**0.6770**	**0.6495**	**0.7743**	**0.7476**	**0.6798**	**0.7969**	**0.7767**	-	-	-



Fig. 7. Visual comparisons of different methods on the DGLCC dataset with 10% labeled rate. (a) RS image. (b) Ground truth. (c) UNetFormer. (d) ST-UNet. (e) MACU-Net. (f) ReCo. (g) U2PL. (h) CAC. (i) CCT. (j) CPS. (k) PSMT. (l) SegMind.

Fig. 9(c) with (e), entropy map with l_e is more concentrated, and the entropy value is generally lower. In Fig. 9(e), high-entropy regions are mainly distributed at the boundaries between different classes. In addition to area boundary, more high-entropy regions in Fig. 9(c) are distributed at area interior compared with Fig. 9(e). Since entropy value can represent prediction uncertainty in some extent, the prediction with l_e is more certainty. Comparing Fig. 9(d) and (f) with (b), semantic segmentation prediction with l_e is generally closer to the ground truth. The result indicates that l_e is conducive to

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TABLE IV
SUMMARY OF THE QUANTITATIVE RESULTS ON THE POTSDAM DATASET

Method	5%			10%			20%			100%		
	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa	mIoU	F1	Kappa
UNetFormer [38]	0.6531	0.7640	0.8036	0.6976	0.7990	0.8395	0.7198	0.8178	0.8525	0.7526	0.8430	0.8758
ST-UNet [39]	0.4950	0.6328	0.6434	0.5421	0.6697	0.6959	0.6074	0.7283	0.7605	0.6930	0.7987	0.8331
MACU-Net [40]	0.3761	0.4810	0.6296	0.4038	0.5997	0.6760	0.4282	0.6235	0.7112	0.6552	0.7632	0.8128
ReCo [46]	0.5755	0.6882	0.7718	0.6203	0.7204	0.8079	0.6776	0.7869	0.8295	-	-	-
U2PL [20]	0.5810	0.6880	0.7661	0.6179	0.7244	0.7895	0.6723	0.7787	0.8279	-	-	-
CAC [47]	0.5279	0.6555	0.7177	0.5594	0.6812	0.7490	0.6237	0.7429	0.7968	-	-	-
CCT [44]	0.4216	0.5206	0.7118	0.5861	0.6872	0.7893	0.6541	0.7698	0.8185	-	-	-
CPS [83]	0.5971	0.7189	0.7702	0.6799	0.7833	0.8355	0.7042	0.8075	0.8452	-	-	-
PSMT [49]	0.3120	0.6308	0.5222	0.5097	0.6282	0.7259	0.6026	0.7229	0.7877	-	-	-
SegMind	**0.6742**	**0.7813**	**0.8162**	**0.7146**	**0.8134**	**0.8470**	**0.7305**	**0.8240**	**0.8597**	-	-	-

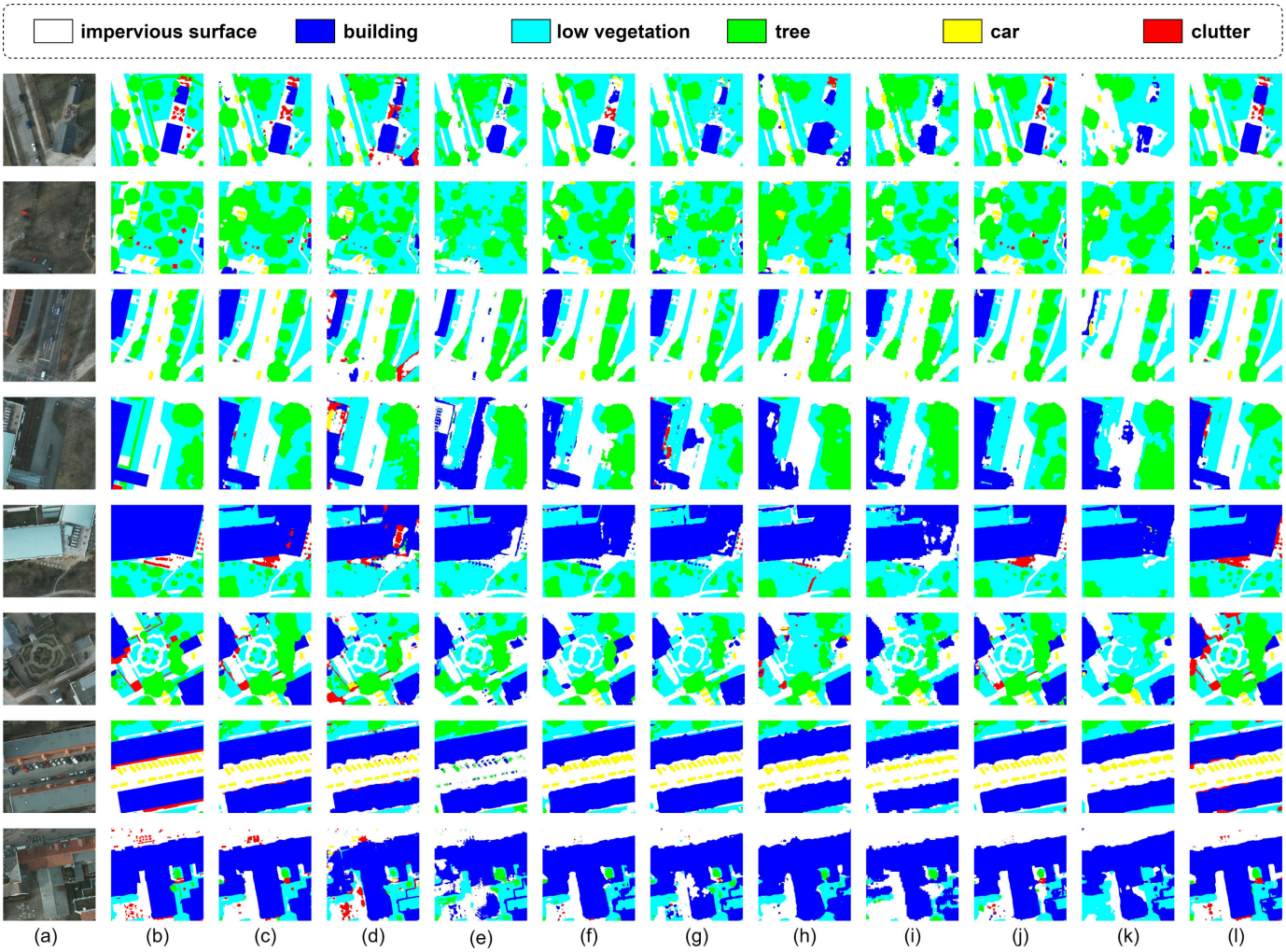


Fig. 8. Visual comparisons of different methods on the Potsdam dataset with 10% labeled rate. (a) RS image. (b) Ground truth. (c) UNetFormer. (d) ST-UNet. (e) MACU-Net. (f) ReCo. (g) U2PL. (h) CAC. (i) CCT. (j) CPS. (k) PSMT. (l) SegMind.

improve network distinguish ability. l_e makes student module output entropy value close to teacher module output entropy value. Since teacher module tends to produce more accurate predictions, l_e tends to reduce student module entropy value, so that it reduces prediction uncertainty. In general, results in Fig. 9 indicate that l_e improves network performance by reducing prediction uncertainty.

Fig. 10 reflects the results of different mask rates on three datasets with 10% labeled rate. As the figure shows, the mIoU values are not monotonically increasing or monotonically

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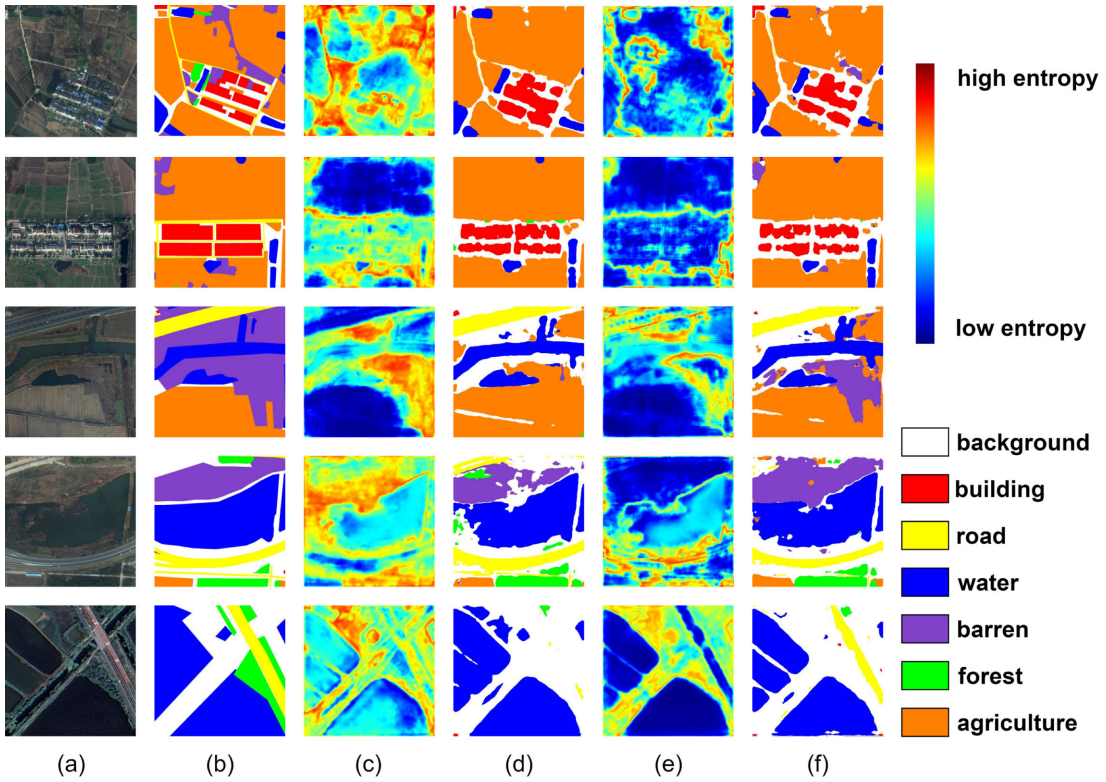


Fig. 9. Results on 10% labeled rate LoveDA dataset with and without l_e. (a) RS image. (b) Ground truth. (c) Entropy map with l_{lab}. (d) Semantic segmentation prediction with l_{lab}. (e) Entropy map with l_{lab} and l_e. (f) Semantic segmentation prediction with l_{lab} and l_e.

TABLE V
MIOU VALUES ON THREE DATASETS WITH DIFFERENT MODULES

MT	MIM	CL	LoveDA	DGLCC	Potsdam
			0.4604	0.5517	0.6673
✔			0.4856	0.6070	0.6851
	✔		0.4763	0.5909	0.6783
		✔	0.4872	0.5822	0.6857
✔	✔	✔	**0.5057**	**0.6495**	**0.7146**

TABLE VI
MIOU VALUES ON THREE DATASETS WITH DIFFERENT LOSS FUNCTIONS

l_{lab}	l_e	l_r	l_{rsc}	l_c	LoveDA	DGLCC	Potsdam
✔					0.4813	0.6157	0.6828
✔	✔				0.4913	0.6287	0.6998
✔		✔			0.4894	0.6282	0.6917
✔			✔		0.4929	0.6247	0.7038
✔				✔	0.4990	0.6237	0.6911
✔	✔	✔	✔	✔	**0.5057**	**0.6495**	**0.7146**

decreasing. Besides, when mask rate is 0.5, the mIoU plunges local minimum in the LoveDA dataset but reaches local maxima in DGLCC and Potsdam datasets. Semantic segmentation is a dense prediction task, mask rate influences image reconstruction and image reconstruction simultaneously, and the experiment is conducted in semisupervised condition; thus,

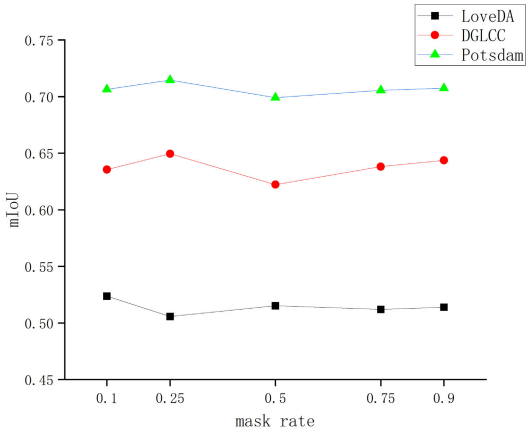


Fig. 10. mIoU values on three datasets with different mask rates.

the mIoU results are not monotonically influenced by the mask rate.

For the sake of exploring the impact of l_r and l_{rsc}, we conduct experiments, and the results are shown in Fig. 11. Notably, the masked area in Fig. 11 caption refers to the area that is masked rather than masked image, i.e., masked area in RS image is $(1-m)\cdot x$ rather than $m\cdot x$. Compared Fig. 11(a) with (c), the visual difference is little. Similar to other MIM reconstruction results, masked area is more blurred compared with original RS image. The areas in Fig. 11(d) are more blurred than areas in Fig. 11(b), but the color and shape in Fig. 11(d) are similar to these in Fig. 11(b). Above observations reflect masked areas loss detailed information but learn general information from unmasked areas by l_r.

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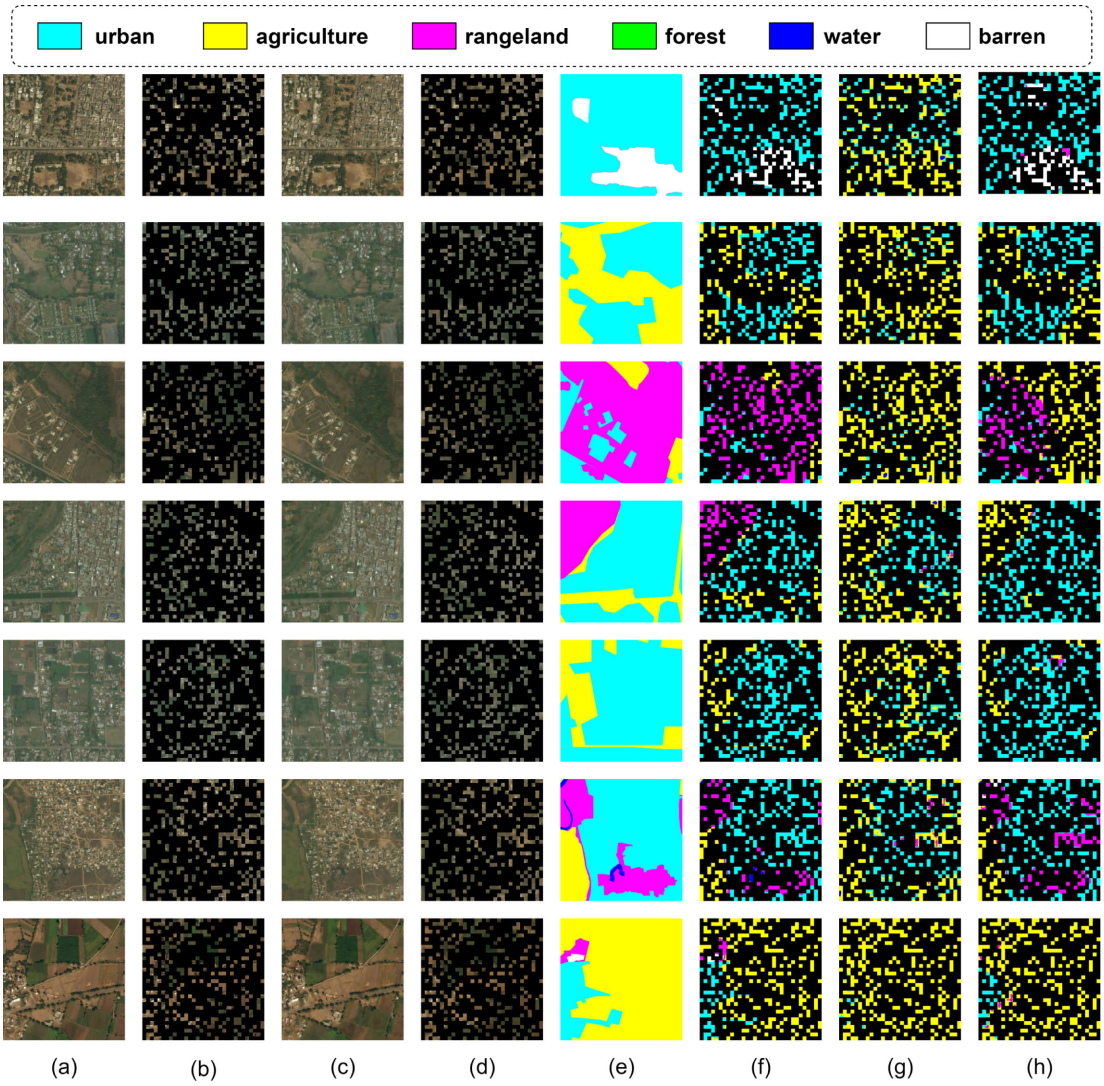


Fig. 11. Reconstruction results and predictions on 10% labeled rate DGLCC dataset. (a) RS image. (b) Masked area in RS image. (c) Combination of unmasked area in RS image and masked area in reconstructed image. (d) Masked area in reconstructed image with l_{lab} and l_r. (e) Ground truth. (f) Masked area in the ground truth. (g) Masked area in prediction with l_{lab} and l_r. (h) Masked area in prediction with l_{lab} and l_{rsc}.

Fig. 11(e) and (f) are the ground truth and masked area in the ground truth, respectively. Compared Fig. 11(g) and (h) with (f), the predictions in Fig. 11(h) are generally more similar to Fig. 11(f). Although l_r learns context information to reconstruct image and lifts context information interactions, l_r does not directly improve feature extraction ability too much. As Table VI shows, the performance with l_{rsc} is higher in LoveDA and Potsdam datasets than performance with l_r. Compared with l_r, l_{rsc} makes predictions in masked area similar to the ground truth, so that it directly lifts context feature extraction ability. Above observations and analyses indicate that l_r lifts context information interactions, and l_{rsc} lifts context information interactions and feature extraction ability.

In order to further illustrate the effect of l_c, we conduct experiments to show the feature distribution with different loss functions. Fig. 12 shows t-distributed stochastic neighbor embedding (t-SNE) 2-D embedding feature distribution in 10% labeled rate LoveDA, DGLCC, and Potsdam datasets. In Fig. 12(a) and (b), points are evenly distributed, and the boundary between different categories is not obvious. In Fig. 12(c), the distribution of points in the same category is more compact, and there are more obvious boundaries between different areas. Besides, the point distance in Fig. 12(c) between the same class is generally closer compared with Fig. 12(a). The difference indicates that l_c lifts network linear separability ability. Notably, some points like "tree" in Fig. 12(c) are obviously more tight, but some points like "barren" in Fig. 12(c) are not significantly closer. The "barren" class is hard to distinguish. Since the pseudo-labels of corresponding class points can be more incorrect, thus, the contrastive loss provides less performance improvement. In the first row, the feature distribution of Fig. 12(d) is slightly more dispersed. In the third row, the "car" features points of Fig. 12(d) are purer than these of Fig. 12(c). The feature distributions in Fig. 12(d) are slightly more distinguishable than these in Fig. 12(c), which indicates that MIM provides additional and helpful perturbations for CL. In general, the feature distributions in Fig. 12(c) and (d) are more distinguishable than these in Fig. 12(a) and (b), which

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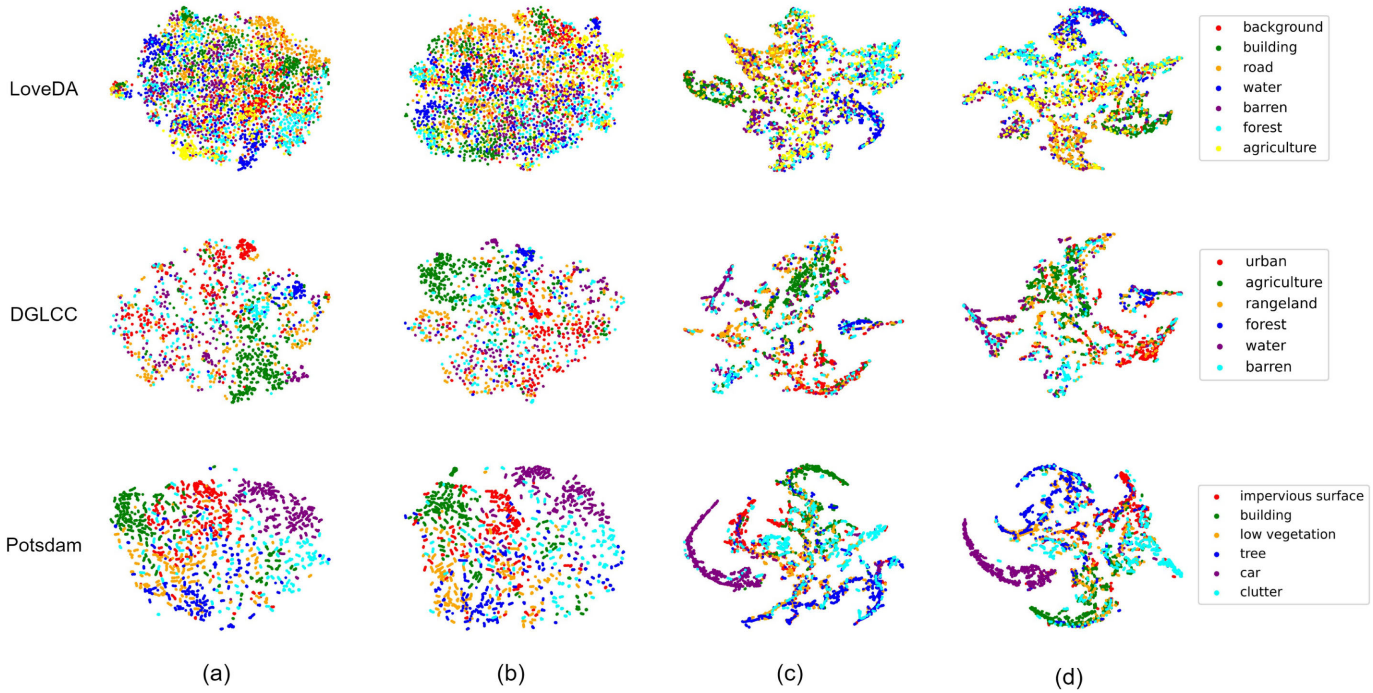


Fig. 12. Visualization of embedding features after applying t-SNE on LoveDA, DGLCC, and Potsdam datasets with 10% labeled rate from SegMind. (a) Feature distribution with MT. (b) Feature distribution with MT and MIM. (c) Feature distribution with MT and CL. (d) Feature distribution with MT, MIM, and CL.

verifies that l_c makes network generate more discriminative features, so that it improves the network performance.

VI. Conclusion

Deep learning-based RS image semantic segmentation has wide applications in many areas, but the lack of annotations has affected its performance. In this article, we proposed a novel semisupervised RS image semantic segmentation network SegMind, which combines MT network with MIM and CL. SegMind employs MIM to enhance the information interactions between different areas of an RS image, thus reducing the dependence of labels. In order to further improve the learned representation linear separability of the proposed model, CL is also introduced to SegMind to distinguish the extracted features from positive/negative samples. The effectiveness of the proposed method has been verified on three widely used datasets. The experimental results show that SegMind gains competitive performances compared with the state-of-the-art methods.

In future research, we will pay attention to other semisupervised learning methods, such as graph neural networks. We also would like to focus on semantic segmentation using 3-D data, in order to get more practical application for our daily life.

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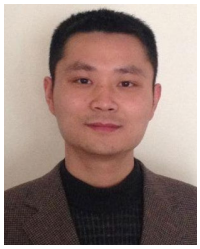
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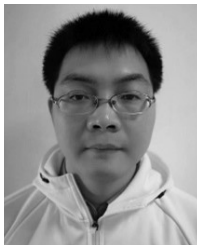
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